Statistics 280 — Practice Midterm 2C — Solutions

Point values per part match the exam. Normal-table values follow the table at the back of the exam. R answers were run in R 4.3 (the ## lines are real output); any equivalent correct R command earns full credit.

Q.1 (30)

a (4 points)
Their own name is 1 of 4 equally likely: $1/4$.

b (8 points)
All $4! = 24$ orderings are equally likely. Count those with nobody in their own position (derangements) by inclusion–exclusion: $24 - \binom{4}{1}3! + \binom{4}{2}2! - \binom{4}{3}1! + \binom{4}{4}0! = 24 - 24 + 12 - 4 + 1 = 9$. So the chance is $9/24 = 3/8 = 0.375$. (Listing works too. Friend 1 draws 2, 3, or 4, and for each choice there are 3 completions.)

c (6 points)
Exactly 2 stuck: choose the 2 who are stuck, $\binom{4}{2} = 6$ ways, and the other two must swap: 6 orderings, chance $6/24 = 1/4$. Exactly 3 stuck is **impossible** (chance 0): if 3 people have their own names, the 4th must too.

d (6 points)
$S = I_1 + I_2 + I_3 + I_4$, where $I_j = 1$ if friend j is stuck. $E(I_j) = 1/4$, so $E(S) = 4 \times \frac{1}{4} = 1$. $E(X + Y) = E(X) + E(Y)$ holds for *any* random variables, dependent or not (Lecture 13). It's variances that need independence. Check from the full distribution: $0 \cdot \frac{9}{24} + 1 \cdot \frac{8}{24} + 2 \cdot \frac{6}{24} + 4 \cdot \frac{1}{24} = 1$.

e (6 points)
Blanks: `1:4`, `draw == 1:4`, `mean(s == 0)`, `mean(s)`.

```
santa = function() {
  draw = sample(1:4)
  return(sum(draw == 1:4))
}
set.seed(6)
s = numeric(100000)
for (i in 1:100000) { s[i] = santa() }
p_none = mean(s == 0); p_none
ES = mean(s); ES
```

```
## [1] 0.37298
## [1] 1.00434
```

Q.2 (30)

a (6 points)
Inclusion–exclusion for three events: $0.55 + 0.40 + 0.30 - 0.20 - 0.15 - 0.10 + 0.05 = 0.85$, so $\mathbf{85\%}$.

b (8 points)
Peel off from the center. Only X and Y: $0.20 - 0.05 = 0.15$. Only X and Z: 0.10. Only Y and Z: 0.05. X only: $0.55 - 0.15 - 0.10 - 0.05 = 0.25$. Y only: $0.40 - 0.15 - 0.05 - 0.05 = 0.15$. Z only: $0.30 - 0.10 - 0.05 - 0.05 = 0.10$. Exactly one: $0.25 + 0.15 + 0.10 = \mathbf{50\%}$. Exactly two: $0.15 + 0.10 + 0.05 = \mathbf{30\%}$. Check: 50% + 30% + 5% (all three) + 15% (none) = 100%.

c (5 points)
$P(Z|X) = 0.15/0.55 \approx 0.27$. This isn't equal to $P(Z) = 0.30$, so they're **not independent**, though they're close.

d (5 points)
Counts out of 1,000: X only 250, Y only 150, Z only 100, X&Y only 150, X&Z only 100, Y&Z only 50, all three 50, and 150 outside all three circles.

e (6 points)
All three: max **30%** (Z inside Y inside X). Min **0%**: $0.55 + 0.40 + 0.30 - 2 < 0$, so no overlap is forced. At least one: min **55%** (Y and Z both inside X), max **100%**: the percentages sum to 125%, so they can cover everyone.

Q.3 (30)

a (5 points)
$0.005(0.95) + 0.995(0.10) = 0.00475 + 0.0995 = 0.10425 \approx 0.10$.

b (6 points)
Prior odds $= 1/199$. Data odds $= 0.95/0.10 = 9.5$. Posterior odds $= 9.5/199 = 0.0477$, so $P = 0.0477/1.0477 \approx \mathbf{0.046}$. Check: $0.00475/0.10425$.

c (6 points)
Each positive test multiplies the odds by 9.5: posterior odds $= 9.5^2/199 = 0.4535$, so $P = 0.4535/1.4535 \approx \mathbf{0.31}$.

d (6 points)
We need odds above 9: $9.5^k/199 > 9$, that is $9.5^k > 1791$. $9.5^3 = 857$ is too small and $9.5^4 = 8145$ is enough, so **4 positives**. Then odds ≈ 40.9 and $P \approx 0.976$; after 3, $P \approx 0.81$.

e (4 points)
Data odds for a negative $= 0.05/0.90 = 0.0556$. Posterior odds $= \frac{1}{199} \times 9.5 \times 0.0556 = 0.00265$, so $P \approx 0.0026$.

f (3 points)
Probably not exactly. Whatever caused a false positive (for example, some other condition) may well cause another, so false positives on repeated tests tend to be correlated. The real evidence from a second positive is then weaker than these calculations suggest.

Q.4 (40)

a (6 points)
$\binom{44}{8} / \binom{50}{8} \approx 0.33$.

```
dhyper(0, 6, 44, 8)
```

```
## [1] 0.3301167
```

b (4 points)

```
1 - phyper(1, 6, 44, 8)
```

```
## [1] 0.2416237
```

c (4 points)

```
phyper(1, 6, 44, 8)
```

```
## [1] 0.7583763
```

d (6 points)
$\binom{8}{1}(0.12)(0.88)^7 = 8 \times 0.12 \times 0.409 \approx 0.39$.

```
dbinom(1, 8, 0.12)
```

```
## [1] 0.3923286
```

e (8 points)
EV $= 500 \times 0.12 = 60$. SE $= \sqrt{500 \times 0.12 \times 0.88} \approx 7.27$. With the continuity correction, use 49.5 to 70.5: $z = \pm 10.5 / 7.27 = \pm 1.45$. The table gives about **85%** between −1.45 and 1.45.

f (6 points)

```
pbinom(70, 500, 0.12) - pbinom(49, 500, 0.12)
pnorm(70.5, 60, sqrt(500 * 0.12 * 0.88)) - pnorm(49.5, 60, sqrt(500 * 0.12 *
0.88))
```

```
## [1] 0.8520698
## [1] 0.8515463
```

g (6 points)
The rule of thumb is $np > 5$ and $n(1-p) > 5$. Part d) has $8 \times 0.12 = 0.96 < 5$, so it **fails**. Part e) has $60 > 5$ and $440 > 5$, so it **passes**. For P(0 of 8): the exact value is $0.88^8 \approx 0.36$. The

normal approximation with EV 0.96 and SE 0.92 gives P(count ≤ 0.5) at $z = -0.5$, about 31%. The histogram is too skewed, and it piles up against 0, so the normal curve doesn't fit it.

Q.5 (30)

a (10 points)

Blanks: `0`, `0`, `total < target`, `total + sample(1:6, 1)`, `n + 1`, `n`.

```
rolls_needed = function(target) {
  total = 0
  n = 0
  while (total < target) {
    total = total + sample(1:6, 1)
    n = n + 1
  }
  return(n)
}
```

b (6 points)

```
set.seed(7)
N = 100000
sims = numeric(N)
for (i in 1:N) { sims[i] = rolls_needed(10) }
mean(sims)                        # E(N)
sd(sims) * sqrt((N - 1) / N)      # SE(N)
mean(sims <= 3)                   # P(N <= 3)
```

```
## [1] 3.32565
## [1] 0.9134233
## [1] 0.62395
```

c (6 points)

$P(N = 1) = 0$, since one roll is at most 6. $N = 2$ exactly when the first two rolls total at least 10: (4,6), (5,5), (5,6), (6,4), (6,5), (6,6). So $P(N = 2) = 6/36 = 1/6$.

d (8 points)

$P(M \le m) = (m/6)^2$, so $P(M = m) = \frac{m^2 - (m-1)^2}{36} = \frac{2m-1}{36}$: 1/36, 3/36, 5/36, 7/36, 9/36, 11/36. $E(M) = \frac{1\cdot1 + 2\cdot3 + 3\cdot5 + 4\cdot7 + 5\cdot9 + 6\cdot11}{36} = \frac{161}{36} \approx 4.47$.

Q.6 (40)

a (6 points)

Average $= 25/5 = 5$. Mean of squares $= (4 + 16 + 16 + 36 + 81)/5 = 30.6$. SD $= \sqrt{30.6 - 25} = \sqrt{5.6} \approx 2.37$.

b (8 points)
Sum: EV $= 25 \times 5 = 125$, SE $= \sqrt{25} \times 2.37 \approx 11.83$. Average: EV $= 5$, SE $= 2.37/\sqrt{25} \approx 0.47$.

c (8 points)
Average ≥ 5.5 means sum ≥ 137.5, so (the sum is a whole number) sum ≥ 138. The continuity correction uses 137.5: $z = (137.5 - 125)/11.83 = 1.06$. The area within 1.06 is about 71.1%, so the chance is about $(100 - 71.1)/2 \approx$ **14.5%**.

d (8 points)
EV $= 3 \times 5 = 15$. That still holds without replacement, because each draw has expected value 5 (Lecture 15). But **no**, the SE formula needs independent draws, which these aren't. Simulate:

```
set.seed(8)
sums = numeric(100000)
for (i in 1:100000) {
  sums[i] = sum(sample(c(2, 4, 4, 6, 9), 3))
}
sd(sums) * sqrt((100000 - 1) / 100000)   # about 2.90; the formula would give
4.10
```

```
## [1] 2.890315
```

e (6 points)
Need $2.366/\sqrt{n} \leq 0.1$, so $\sqrt{n} \geq 23.66$, so $n \geq 5.6/0.01 =$ **560**.

f (4 points)
Four times as many draws: the SE of the sum doubles ($\times\sqrt{4}$) to about 23.7, and the SE of the average halves to about 0.24. That's the law of averages again.

Q.7 (15 extra credit)

Let $\mu_X = E(X)$ and $\mu_Y = E(Y)$, which are constants. Expand and use linearity:

$$E[(X - \mu_X)(Y - \mu_Y)] = E(XY) - \mu_Y E(X) - \mu_X E(Y) + \mu_X \mu_Y = E(XY) - \mu_X \mu_Y.$$

Then $\text{Cov}(X + Y, X - Y) = E[(X + Y)(X - Y)] - E(X + Y)E(X - Y) = E(X^2) - E(Y^2) - (\mu_X^2 - \mu_Y^2) = \text{Var}(X) - \text{Var}(Y) = 0$.